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Education

Peking University — Beijing, China
Ph.D. in Computer Science — Sept. 2023 – Present

Peking University — Beijing, China
B.Sc in Computer Science & Mathematics — Sept. 2019 – Jun. 2023

Working Experience

Alibaba Group — Beijing, China
Research Intern, LLM Inference — Oct. 2025 – Present

ByteDance — Beijing, China
Research Intern, Network Infrastructure — Aug. 2025 – Oct. 2025

Infinigence AI — Beijing, China
Research Intern, System Communication — Dec. 2024 – Apr. 2025

Baichuan AI — Beijing, China
Research Intern, LLM Inference — Apr. 2024 – Dec. 2024

Theory Lab, Huawei Technologies — Hong Kong SAR, China
Research Intern (Remote), LLM Training — Sept. 2023 – Apr. 2024

Publications

HourglassSketch: An Efficient and Scalable Framework for Graph Stream Summarization
Jiarui Guo, *Boxuan Chen, Kaicheng Yang, Tong Yang, Zirui Liu, Qiuheng Yin, Sha Wang, Yuhan Wu, Xiaolin Wang, Bin Cui, Tao Li, Xi Peng, Renhai Chen, Gong Zhang*
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CuckooGraph: A Scalable and Space-Time Efficient Data Structure for Large-Scale Dynamic Graphs
Zhuochen Fan, Yalun Cai, Zirui Liu, ***Jiarui Guo****, Xin Fan, Tong Yang, Bin Cui*
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SteadySketch: A High-Performance Algorithm for Finding Steady Flows in Data Streams
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Enabling Space-Time Efficient Range Queries with REncoder
Zhuochen Fan, Bowen Ye, Ziwei Wang, Zheng Zhong, ***Jiarui Guo****, Yuhan Wu, Haoyu Li, Tong Yang, Yaofeng Tu, Zirui Liu, Bin Cui*
International Journal on Very Large Data Bases (**VLDBJ**). Aug. 2024

SketchPolymer: Estimate Per-item Tail Quantile Using One Sketch
Jiarui Guo*, Yisen Hong, Yuhan Wu, Yunfei Liu, Tong Yang, Bin Cui*
2023 ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD '23**). Aug. 2023

SteadySketch: Finding Steady Flows in Data Streams
Xiaodong Li, Zhuochen Fan, Haoyu Li, Zheng Zhong, ***Jiarui Guo****, Sheng Long, Tong Yang, Bin Cui*
2023 IEEE/ACM International Symposium on Quality of Service (**IWQoS '23**). Jun. 2023

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Finding Simplex Items in Data Streams
*Zhuochen Fan, **Jiarui Guo**, Xiaodong Li, Tong Yang, Yikai Zhao, Yuhan Wu, Bin Cui, Yanwei Xu, Steve Uhlig, Gong Zhang*
2023 IEEE International Conference on Data Engineering (**ICDE '23**). Apr. 2023

REncoder: A Space-Time Efficient Range Filter with Local Encoder
*Ziwei Wang, Zheng Zhong, **Jiarui Guo**, Yuhan Wu, Haoyu Li, Tong Yang, Yaofeng Tu, Huanchen Zhang, Bin Cui*
2023 IEEE International Conference on Data Engineering (**ICDE '23**). Apr. 2023

BurstSketch: Finding Bursts in Data Streams
*Ruijie Miao, Zheng Zhong, **Jiarui Guo**, Zikun Li, Tong Yang, Bin Cui*
IEEE Transactions on Knowledge and Data Engineering (**TKDE**). Nov. 2022

PISketch: Finding Persistent and Infrequent Flows
*Zhuochen Fan, Zhoujing Hu, Yuhan Wu, **Jiarui Guo**, Wenrui Liu, Tong Yang, Hengrui Wang, Yifei Xu, Steve Uhlig, Yaofeng Tu*
2022 ACM SIGCOMM Workshop on Formal Foundations and Security of Programmable Network Infrastructures (**FFSPIN '22**). Aug. 2022